

1.9.11

EE25BTECH11022 - sankeerthan

Question: if the distance between the points $(k, -2)$ and $(3, -6)$ is 10 units, find the positive value of k .

solution: Let the given points be

$$\mathbf{A} = \begin{pmatrix} k \\ -2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 3 \\ -6 \end{pmatrix} \quad (0.1)$$

The direction vector of the segment joining $\mathbf{A}$ and $\mathbf{B}$ is given by:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 3 - k \\ -6 - (-2) \end{pmatrix} = \begin{pmatrix} 3 - k \\ -4 \end{pmatrix} \quad (0.2)$$

The length of the segment is the magnitude of the direction vector:

$$|\mathbf{B} - \mathbf{A}| = \begin{pmatrix} 3 - k \\ -6 - (-2) \end{pmatrix} = \begin{pmatrix} 3 - k \\ -4 \end{pmatrix} \quad (0.3)$$

The distance between points $\mathbf{A}$ and $\mathbf{B}$ is given as, $d = \|\mathbf{B} - \mathbf{A}\| = 10$

$$\|\mathbf{B} - \mathbf{A}\| = \sqrt{(\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A})} \quad (0.4)$$

$$(\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A}) = \|\mathbf{B} - \mathbf{A}\|^2 \quad (0.5)$$

$$(\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A}) = (10)^2 \quad (0.6)$$

$$100 = (3 - k \quad -4) \begin{pmatrix} 3 - k \\ -4 \end{pmatrix} \quad (0.7)$$

$$100 = (3 - k) \times (3 - k) + (-4) \times (-4) \quad (0.8)$$

$$100 = (3 - k)^2 + 16 \quad (0.9)$$

$$(3 - k)^2 = 84 \quad (0.10)$$

$$3 - k = \pm \sqrt{84} \quad (0.11)$$

$$k = 3 + \sqrt{84}, 3 - \sqrt{84} \quad (0.12)$$

Therefore, the positive value of k is $3 + \sqrt{84} \approx 12.17$

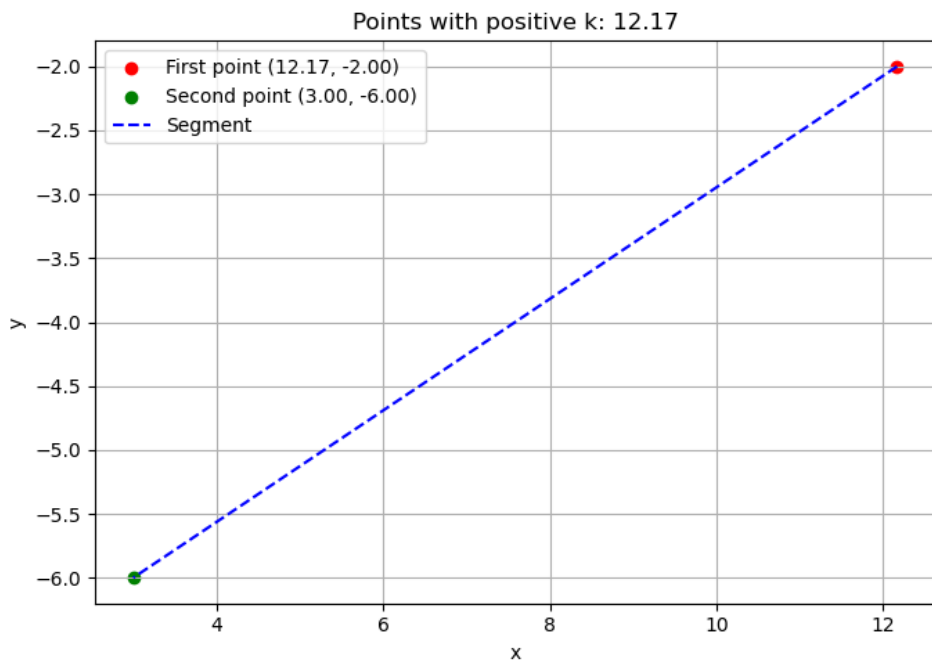


Fig. 0.1: Plot of line segment **AB**